

Office Automation Software Lab

Code: 1259

Semester: 1

Credits: 1.5

Core Course

Mastering MS Word, Excel, PowerPoint, and Access for Business Professionals.

Official Source Declaration

[illegible]

This lesson is architected using the official **SITTTR Kerala Syllabus (Revision 2021)** for Course Code **1259**. All modules, contact hours, and learning outcomes are mapped directly from the official document.

Course Dashboard

Program	Diploma in Computer Application and Business Management
Course Category	Program Core
Periods per Week	3 (Practical)
Total Periods	45
Prerequisites	Basic Computer Knowledge (Secondary School level)

How to Study this Lab Course

- **Hands-on Practice:** This is a lab course. Every topic mentioned must be practiced on a computer.
- **Step-by-Step Documentation:** Maintain a lab record for every experiment.
- **Business Context:** Always think how these tools apply to a business office (e.g., creating a salary slip in Excel).

Course Outcomes (CO) & Mapping

CO	Description	Hours	Cognitive Level
CO1	Use Word Processing for documents and reports	8	Applying
CO2	Create basic spreadsheets for business	10	Applying
CO3	Apply Advanced Excel formulas and manipulation	12	Applying
CO4	Build PPT presentations and MS Access databases	12	Applying

CO-PO Mapping

Strongly mapped (3) to PO1 (Basic Knowledge) and PO4 (Engineering Tools).

Syllabus Coverage Map

Module	Key Topics	Weightage
1	Word: Mail Merge, Tables, PDF Export	18%
2	Excel: Data Referencing, Logical IF, Auto Sum	22%
3	Advanced Excel: VLOOKUP, Pivot Tables, Dashboards	27%
4	PPT & Access: Business Slides, Database Queries	27%
Exam	Lab Exam I & II	6%

Learning Roadmap

Step 1: Master Document Formatting (Word) → **Step 2:** Learn Data Calculation (Excel Basic) → **Step 3:** Professional Data Analysis (Excel Advanced) → **Step 4:** Data Presentation & Storage (PPT/Access).

Module 1: Word Processing Applications (CO1)

Duration: 8 Hours Focus: Professional Reporting

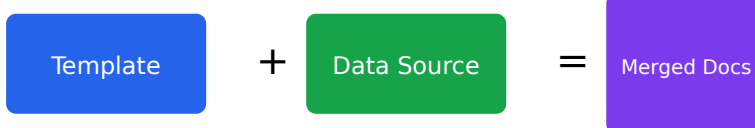
M1.01: Document Creation & Formatting

Creating professional documents involves setting margins, font styles, paragraph spacing, and headers/footers.

Malayalam support: മലയാളം സാക്ഷരതയും കമ്പ്യൂട്ടറിലെ അക്ഷരങ്ങൾ ഉപയോഗിച്ച് പ്രൊഫഷണൽ ഡോക്യുമെന്റുകൾ സൃഷ്ടിക്കുകയും ചെയ്യേണ്ടതാണ്.

M1.02: Mail Merge for Multiple Letters

Mail Merge is used to create personalized letters for hundreds of recipients using a single template and a data source (like an Excel list).



M1.03: Tables, Hyperlinks & PDF Export

- **Tables:** Insert > Table. Adjust borders and shading.
- **Hyperlinks:** Ctrl+K to link text to websites or other document sections.
- **Export:** File > Export > Create PDF/XPS.

Module 2: Basic Spreadsheets (CO2)

Duration: 10 Hours Focus: Business Calculations

M2.02: Rows, Columns & Referencing

Understanding **Relative** (A1) vs **Absolute** (\$A\$1) referencing is crucial for copying formulas.

Malayalam support: ഒരു കോശം കോപ്പി ചെയ്തപ്പോൾ അത് അടുത്തുള്ള കോശങ്ങളിലേക്ക് മാത്രമാണ് കോപ്പി ചെയ്യുന്നത്. ഇത് ആപേക്ഷിക പരാമർശനം (Relative Reference) ആണ്. അത് കോപ്പി ചെയ്തപ്പോൾ കോശത്തിന്റെ സ്ഥാനം മാറുന്നു. ഉദാഹരണം: A1 കോശത്തിൽ =A2+1 എന്ന ഫോർമുല ഉൾക്കൊള്ളുന്നു. ഇത് B1 കോശത്തിലേക്ക് കോപ്പി ചെയ്താൽ, ഫോർമുല =B2+1 ആയി മാറുന്നു.

M2.03: Basic Formulae & IF Logic

Scenario: Calculate Pass/Fail based on marks (Pass mark = 40).
Formula: =IF(B2>=40, "Pass", "Fail")
Logic: If value in B2 is 40 or more, display "Pass", else "Fail".

Module 3: Advanced Excel (CO3)

M3.01: Charts & Lookup Functions

VLOOKUP: Used to search for a value in the first column of a table and return a value in the same row from another column.

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

M3.03: Pivot Tables & Data Protection

Pivot Tables: Summarize large datasets instantly (e.g., total sales per region).

Protection: Review > Protect Sheet. Prevents unauthorized editing of formulas.

M3.04: Basic Dashboard Preparation

A dashboard combines charts, slicers, and key metrics into a single screen for business decision-making.

Module 4: Presentations & Databases (CO4)

M4.01: Professional PowerPoint

- Use **Slide Master** for consistent branding.
- Insert Charts directly from Excel to keep data linked.
- Use **SmartArt** for process flows.

M4.02 & M4.03: MS Access Basics

MS Access is a Relational Database Management System (RDBMS).

Term	Description
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Table	Stores data in rows (records) and columns (fields).
Query	A request to retrieve specific data (e.g., List all students from 'Kochi').
Primary Key	A unique ID for each record (e.g., Admission Number).

Official Lab Activities

Activity	Description
Resume Preparation	Create a 2-page professional resume using Word tables and formatting.
Salary Sheet	Create an automated payroll in Excel using IF, SUM, and HRA/DA percentages.
Sales Dashboard	Pivot table analysis of a yearly sales dataset.
Student DB	Create an Access database for student marks and query the top performers.

Formula & Procedure Bank

Excel: SUMIF =SUMIF(Range, Criteria, Sum_Range) <i>Use: Sum values based on a condition.</i>
Excel: CONCATENATE =A2 & " " & B2 <i>Use: Join First Name and Last Name.</i>

Visual Revision Library

Excel Cell Anatomy

Access Relationship
Table A (PK) ∞---1 Table B (FK)

Assessment Methodology

Type	Details	Weight
Lab Exam I	Covers Module 1 & 2 (Word & Basic Excel)	1.5 Hours
Lab Exam II	Covers Module 3 & 4 (Adv Excel, PPT, Access)	1.5 Hours
Open Ended	Creative project combining multiple tools	3 Hours

Module-wise Q&A

Q1: What is the main advantage of Mail Merge? (CO1)

It allows the mass production of personalized letters or labels quickly by linking a template to a database, saving time and reducing manual errors.

Q2: Explain the difference between COUNT and COUNTA in Excel. (CO2)

COUNT only counts cells containing numbers. COUNTA counts all non-empty cells (numbers, text, errors, etc.).

Q3: What is a 'Slicer' in an Excel Dashboard? (CO3)

A Slicer is a visual filtering tool used with Pivot Tables and Dashboards to filter data with a single click.

Q4: What is a Primary Key in MS Access? (CO4)

A Primary Key is a field that uniquely identifies each record in a table. It cannot contain null values or duplicate data.

Practice Model Paper

(Not an official examination paper)

Part A (Word - 10 Marks): Create a formal meeting notice. Use Mail Merge to address it to 5 different department heads listed in an Excel sheet.

Part B (Excel - 20 Marks): Create a table with 'Product Name', 'Price', and 'Quantity'. Calculate 'Total'. Use an IF statement to apply a 10% discount if Total > 5000.

Part C (Access - 10 Marks): Create a table 'Employees'. Fields: EmpID (PK), Name, Salary. Insert 5 records. Write a query to show employees with Salary > 25000.

Quick Revision Checklist

- ☒ Can I perform a Mail Merge?
- ☒ Do I know the difference between \$A1, A\$1, and \$A\$1?
- ☒ Can I create a VLOOKUP without errors?
- ☒ Can I create a Pivot Table and add a Chart to it?
- ☒ Do I know how to set a Primary Key in Access?

Glossary

Ribbon

The tabbed toolbar at the top of Office applications.

Workbook

An Excel file containing one or more worksheets.

Query

A specific search or data retrieval command in a database.

Transition

The visual effect that occurs when moving from one slide to another in PPT.

Official References

- R1: Bittu Kumar, Mastering MS Office, V&S Publishers, 2017.
- R2: Michael Price, Office 2019 in Easy Steps, BPB Publications, 2019.
- Online: